KNRT MARKETPLACE

Technical Whitepaper

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KEY FEATURES

- Decentralized Marketplace
- Cryptographic Identity Management
- Multi-Signature Escrow Protocol
- Hybrid Reputation System
- Legal Framework Integration
- Dispute Resolution Mechanism
- Cross-Platform Implementation

Document Classification: Technical Specification
Document Type: Whitepaper
Intended Audience: Developers, Architects, Security Researchers

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1. EXECUTIVE SUMMARY

KNRT Marketplace implements a peer-to-peer NFT marketplace with cryptographic escrow and reputation tracking. The system combines Ethereum Layer 2 settlement, IPFS content addressing, and Ed25519 cryptographic identity.

Key Innovations

- Zero-Custody Escrow: Multi-signature escrow where neither party nor platform controls funds unilaterally
- Cryptographic Reputation: Ed25519-signed reputation records with Merkle tree aggregation
- Hybrid Storage Model: Critical data on-chain, bulk data on IPFS with integrity verification
- Event-Driven Architecture: Microservices communicating via Redis Streams
- Legal Framework Integration: Jurisdiction-aware dispute resolution with evidence preservation

Target Audience

- Developers building decentralized marketplaces
- Security researchers evaluating cryptographic implementations
- Legal professionals assessing jurisdiction-aware dispute resolution
- Enterprises considering blockchain-based commerce infrastructure

2. INTRODUCTION TO KNRT MARKETPLACE

2.1 Platform Overview

KNRT Marketplace is a decentralized marketplace for digital assets and services. The platform supports:

- Digital Assets: NFTs, tokenized documents, software licenses
- Services: Freelance work, consulting, custom development
- Transaction Types: Fixed price, auctions, escrow-protected trades

The platform operates without custody of user funds. Smart contracts enforce escrow conditions, IPFS stores metadata with content addressing, and cryptographic signatures provide identity verification.

2.2 Design Principles

Trust Minimization

Trust is distributed across:

- Smart contracts (Ethereum Layer 2)
- Cryptographic primitives (Ed25519, SHA-256)
- Decentralized storage (IPFS)
- Reputation aggregators (multiple independent operators)

No single entity controls all three.

Cost Minimization

Ethereum Layer 2 (Optimism) reduces transaction costs by approximately 90% compared to Ethereum mainnet. Typical transaction costs:

- Mint: $0.01–$0.05
- Trade: $0.02–$0.10
- Escrow: $0.05–$0.20

Data Ownership

- Identity: Ed25519 keypairs generated locally, never transmitted
- Reputation: Signed records exportable to any compatible system
- Content: IPFS addressing enables retrieval from any node

Jurisdiction Awareness

- Dispute resolution supports multiple legal frameworks
- Evidence preservation meets cross-jurisdiction requirements
- Platform does not assume single-jurisdiction operation

3. TECHNICAL ARCHITECTURE

3.1 Zero-Layer 2 Network

Smart contracts deploy on Optimism, an Ethereum Layer 2 network. Transactions inherit Ethereum's security while achieving approximately 90% cost reduction.

Why Not Mainnet?

At 30 gwei gas price and $3,000 ETH:

- Simple transfer: $0.30
- Contract interaction: $3.00–$15.00
- Complex escrow operation: $15.00–$50.00

Layer 2 reduces these to $0.01–$0.50 while maintaining Ethereum security guarantees.

Why Not Other L2s?

Optimism provides:

- Mature tooling (Hardhat, OpenZeppelin)
- Active development ecosystem
- Clear path to Ethereum mainnet if needed
- Compatibility with standard Ethereum wallets

3.2 Event-Driven Microservices

All backend services communicate via Redis Streams. Services subscribe to relevant event streams and maintain local state. Failure in one service does not cascade to others.

```
Blockchain Indexer → Redis Streams → Reputation Aggregator
                   → Redis Streams → Notification Service
                   → Redis Streams → Analytics Service
```

Benefits

- Decoupled deployment: Services can be updated independently
- Fault tolerance: Failed consumers do not affect producers
- Scalability: Add consumers without modifying producers
- Observability: Event streams provide audit trail

Implementation

- Runtime: Bun 1.0+ (TypeScript)
- Framework: Custom event handlers
- Persistence: Redis Streams with consumer groups

3.3 Data Ownership Model

Data	Storage	Ownership
Identity	Local keystore	User controls private key
Reputation	Signed records	Exportable, portable
Content	IPFS	Content addressing
Transactions	Blockchain	Immutable record

Identity

Ed25519 keypairs generated locally. Private key never transmitted. Signature verification proves ownership without revealing private key.

Reputation

Reputation records are signed by aggregators and can be exported and verified by any compatible system. Multiple aggregators prevent single-point manipulation.

Content

Content addressing (IPFS CIDs) ensures retrieval from any node. Platform cannot modify stored content.

Transactions

Blockchain provides immutable record of escrow events. Dispute resolution can reference this record.

4. CORE FEATURES & FUNCTIONALITY

4.1 NFT Creation & Minting

Users create NFTs by uploading metadata to IPFS and calling the marketplace contract. Supported standards: ERC-721 (single item), ERC-1155 (batch).

Metadata Structure

```
{
  "name": "Example Asset",
  "description": "Description",
  "image": "ipfs://Qm...",
  "attributes": [
    {"trait_type": "Type", "value": "Example"}
  ]
}
```

Minting Flow

1. User uploads metadata to IPFS, receives CID
2. User calls mint(to, tokenURI) on contract
3. Contract emits Minted(tokenId, to, tokenURI)
4. Indexer detects event, updates database

4.2 Marketplace Trading

Fixed Price

- Seller lists at specified price in USDC
- Buyer approves USDC spending
- Buyer calls buyFixed(listingId)
- Contract transfers NFT to buyer, USDC minus fee to seller

Escrow

- Buyer creates escrow with seller, amount, deadline
- Buyer funds escrow
- Seller delivers off-chain
- Buyer confirms receipt
- Contract releases funds

Auctions

- Seller creates auction with reserve price and duration
- Bidders place bids (USDC held in escrow)
- Anti-snipe: bids in final 5 minutes extend auction by 5 minutes
- Winner calls settleAuction() after auction ends
- Losing bids refunded

4.3 Reputation System

Ed25519 Signature Verification

```
// Reputation record structure
interface ReputationRecord {
  subject: PublicKey,
  score: number,
  timestamp: Unix,
  evidence: [TxHash],
  aggregator: PublicKey,
  signature: Ed25519Sig
}

// Verification
function verify(record, aggregatorPubkey):
  return ed25519.verify(
    record.signature,
    hash(record.subject || record.score || record.timestamp || record.evidence),
    aggregatorPubkey
  )
```

Merkle Tree Aggregation

Aggregators collect reputation records and publish Merkle roots periodically. Individual records can be verified against the root without downloading the full tree.

Sybil Resistance

- Minimum transaction count before reputation accrues
- Weight by counterparty reputation
- Time decay: recent transactions weight more
- Dispute penalties: losing disputes reduces score

5. DATA ARCHITECTURE

5.1 Hybrid Storage Model

Data	Storage	Rationale
Escrow state	Blockchain	Must be immutable for trustless execution
Reputation roots	Blockchain	Anchor for verification
Reputation records	Off-chain	Too expensive for full storage
Metadata	IPFS	Content addressing ensures integrity
Images	IPFS	Bandwidth cost prohibitive on chain
Indexes	Database	Rebuilds from blockchain events

Principle

Blockchain provides ordering and finality. Off-chain systems provide speed and storage. Cryptography connects them without requiring trust.

5.2 Data Schemas

Escrow

```
struct Escrow {
  address buyer;
  address seller;
  address token;
  uint256 amount;
  uint256 deadline;
  bytes32 metadataHash;
  EscrowStatus status;
}
```

Listing

```
struct Listing {
  address seller;
  address nftContract;
  uint256 tokenId;
  address paymentToken;
  uint256 price;
  uint256 expiry;
  ListingStatus status;
}
```

Auction

```
struct Auction {
  address seller;
  address nftContract;
  uint256 tokenId;
  address paymentToken;
  uint256 reserve;
  uint256 startTime;
  uint256 endTime;
  address highBidder;
  uint256 highBid;
  bool settled;
}
```

5.3 Cryptographic Integrity

Content Addressing

IPFS CIDs derived from content hash. Modification changes CID.

Merkle Proofs

Reputation roots allow inclusion proofs. Prove record exists without revealing other records.

Signature Chains

Reputation records include evidence transaction hashes. Disputes can trace back to blockchain events.

6. SECURITY MODEL

6.1 Threat Model

Adversary Capabilities

- Can read all blockchain data
- Can submit transactions
- Can operate multiple identities
- Can attempt MEV extraction
- Can compromise individual keys
- Can attempt 51% attack on Layer 2

Attack Vectors

Attack	Mitigation
Reentrancy	Checks-effects-interactions pattern
Front-running	Commit-reveal where applicable
Sybil	Minimum transaction count, weight by counterparty reputation
Oracle manipulation	Multi-source aggregation
Key compromise	Multi-sig for critical operations
Dispute manipulation	Evidence preservation, multi-jurisdiction support

Out of Scope

- Quantum cryptanalysis
- Social engineering
- Physical key extraction
- Regulatory changes

6.2 Cryptographic Primitives

Purpose	Algorithm
Identity	Ed25519
Hashing	SHA-256
Merkle trees	Keccak-256
Encryption	AES-256-GCM
Key derivation	HKDF-SHA256

Ed25519

- Signature size: 64 bytes
- Public key size: 32 bytes
- Deterministic signatures (no nonce reuse vulnerability)
- Fast verification

Commit-Reveal

```
commitment = hash(value || nonce)
// Later:
verify(commitment, value, nonce)
```

6.3 Key Management

Identity Keys

- Generated locally from secure random
- Private key never transmitted
- Backup: encrypted export with passphrase

Escrow Keys

- Multi-sig: 2-of-3 (buyer, seller, platform)
- Platform key: hardware security module
- Key ceremony: documented, witnessed

Upgrade Keys

- Multi-sig: 3-of-5
- Timelock: 48 hours minimum
- Public announcement before execution

7. ECONOMIC MODEL

7.1 Fee Structure

Transaction	Fee
Fixed price	2.5%
Escrow	2.5%
Auction	2.5%
Dispute resolution	5%

Distribution

- Protocol treasury: 1.5%
- Reputation aggregators: 0.5%
- Dispute resolvers: 0.5%

7.2 Escrow Economics

Gas Costs

Operation	Cost
Create escrow	$0.05–$0.20
Release	$0.02–$0.10
Dispute	$0.10–$0.50

Minimum Viable Transaction

Escrow becomes economically viable when transaction value exceeds approximately $10.

7.3 Incentive Alignment

Buyers

- Escrow protection
- Dispute resolution
- Reputation accumulation

Sellers

- Escrow guarantee
- Reputation accumulation
- Dispute resolution

Aggregators

- Fee share
- Reputation for accurate aggregation

Protocol

- Sustainable fee revenue
- Ecosystem growth

8. LEGAL FRAMEWORK

8.1 Jurisdiction

Platform Position

- Software provider, not marketplace operator
- Does not custody user funds
- Does not mediate disputes
- Does not verify identities

Implications

- Platform liability limited
- Users responsible for own compliance
- Dispute resolution optional
- Evidence preservation supports any jurisdiction

8.2 Dispute Resolution

Evidence Types

- Transaction hashes
- Signature verifications
- Timestamps
- Communication logs

Resolution Options

- Bilateral negotiation
- Arbitration
- Court proceedings

8.3 Evidence Preservation

Retention

- Blockchain: permanent
- IPFS: pinned while relevant
- Off-chain: 7 years recommended

Integrity

- Hashes prove content unchanged
- Signatures prove authorship
- Merkle proofs prove inclusion

9. IMPLEMENTATION

9.1 Technology Stack

Component	Technology
Runtime	Bun 1.0+
Language	TypeScript
Framework	Custom event handlers
Database	PostgreSQL
Cache	Redis
Streams	Redis Streams
Blockchain	Ethereum Optimism
Contracts	Hardhat, OpenZeppelin
Storage	IPFS

9.2 Deployment Architecture

```
Load Balancer
  → Gateway Instance 1
  → Gateway Instance 2
  → Gateway Instance 3

Indexer (Single Instance)

Reputation Aggregator (Single Instance)

PostgreSQL (Primary + Replicas)

Redis (Cluster)

IPFS Node
```

9.3 Monitoring

Metrics

- Transaction volume
- Escrow state distribution
- Dispute rate
- Indexer lag
- Error rate

Alerts

- Indexer lag > 100 blocks
- Error rate > 1%
- Dispute spike
- Contract anomaly

10. ROADMAP

10.1 Phase 1: Foundation (Q1 2026)

- Core contracts deployed
- Basic indexer operational
- Fixed price trading
- Escrow trading
- Ed25519 reputation records
- Merkle tree aggregation

10.2 Phase 2: Ecosystem (Q2 2026)

- Auction contracts
- Multi-aggregator support
- Dispute resolution
- Mobile wallet integration
- Developer documentation

10.3 Phase 3: Governance (Q3 2026)

- Protocol governance
- Fee adjustment
- Aggregator selection
- Upgrade proposals

10.4 Phase 4: Decentralization (Q4 2026)

- Decentralized indexer
- Decentralized aggregation
- Cross-chain support
- Protocol autonomy

11. CONCLUSION

11.1 Summary

KNRT Marketplace implements a peer-to-peer NFT marketplace with:

- Zero-custody escrow
- Cryptographic reputation
- Hybrid storage
- Event-driven architecture
- Legal framework integration

11.2 Vision

Commerce infrastructure should:

- Minimize trust requirements
- Preserve user ownership
- Support jurisdiction diversity
- Enable composability

11.3 Call to Action

Developers

- Review contracts
- Run indexer
- Integrate API

Security Researchers

- Audit contracts
- Review cryptography
- Test edge cases

Legal Professionals

- Review jurisdiction handling
- Assess evidence preservation
- Provide feedback

12. APPENDIX

12.1 Glossary

Term	Definition
Escrow	Third party holds funds until conditions met
Multi-sig	Multiple signatures required
Merkle tree	Hash tree enabling inclusion proofs
Commit-reveal	Two-phase protocol hiding intent
Sybil	Fake identities
Front-running	Exploiting transaction ordering
Reentrancy	Recursive contract exploit
Layer 2	Scaling solution inheriting Ethereum security
Optimism	Specific Layer 2 implementation
Ed25519	Elliptic curve signature algorithm
IPFS	Content-addressed storage network
CID	Content identifier
Indexer	Service processing blockchain events
Aggregator	Service computing reputation scores
Gateway	Service handling user requests

12.2 References

Standards

- ERC-721: Non-fungible token standard
- ERC-1155: Multi-token standard
- EIP-712: Typed structured data hashing and signing
- RFC 8032: Edwards-curve Digital Signature Algorithm (Ed25519)

Documentation

- Optimism: https://docs.optimism.io
- OpenZeppelin: https://docs.openzeppelin.com
- IPFS: https://docs.ipfs.tech
- Merkle Proofs: https://en.wikipedia.org/wiki/Merkle_tree

Research

- Ethereum Foundation Blog
- Vitalik Buterin's blog
- Various DeFi security analyses

12.3 Contact

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Contributions Welcome

- Technical review
- Security analysis
- Implementation feedback
- Documentation improvements

Feedback Channels

- GitHub Issues
- Pull Requests
- Direct Contact

Welcome to the Future of Digital Ownership

Commerce infrastructure should minimize trust requirements, preserve user ownership, support jurisdiction diversity, and enable composability. This whitepaper describes one implementation of these principles.

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